

UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

INITIAL COMMENTS OF ON ENERGY STORAGE, INC.

ON Energy Storage, Inc. (ON) respectfully submits these initial comments on the proposed advance notice of proposed rulemaking submitted by the Secretary of Energy to the Federal Energy Regulatory Commission (Commission).¹ ON strongly supports the effort by the Commission to consider potential reforms to large load interconnection. These reforms should ensure the United States has an interconnection process that enables us to remain a market leader in AI and advanced computing, while also ensuring our grid remains reliable, stable, and affordable for existing users. North American Electric Reliability Corporation's (NERC) issuance of a Level 3 alert on large computational loads highlights the risks these large loads can introduce to the grid.²

ON designs, builds, and operates battery storage and power systems for AI data centers and other mission-critical facilities. Our flagship product, the AI UPSTM, is the first medium-voltage uninterruptible power supply (MV UPS) purpose-built for grid-safe AI data centers. It sits between generation (behind the meter or grid) and load, consisting of a grid-following inverter on the generation side and a grid-forming inverter on the load side. It converts what would otherwise be a volatile graphics processing unit (GPU) load into a deterministic, ramp-controlled, ride-through-compliant resource at the meter that behaves comparably to or better than existing loads.

¹ *Notice Inviting Comments*, Docket No. RM26-4-000 (Oct. 27, 2025)

² NERC, *Essential Action to Industry: Computational Load Modeling, Studies, Instrumentation, Commissioning, Operations, Protection, and Control*, Level 3 Alert, Initial Distribution May 4, 2026, available at www.nerc.com/globalassets/programs/bpsa/alerts/level-3-computational-load-alert.pdf

We have 3 GW of AI UPSTM under construction with commissioning beginning this summer and deliveries extending through 2026 and into 2027.

FERC and state policymakers have focused on the capacity expansion costs and challenges associated with large computational loads. These risks may result in widespread blackouts or significantly higher costs for existing customers. We support efforts to resolve these issues in an equitable manner, while ensuring we do not hold back progress on artificial intelligence. Often overlooked are the power quality and system stability risks posed by these loads that are larger, more volatile, and more fragile than what grid operators are accustomed to seeing. We have already seen some of these risks on the grid today. On July 10, 2024, a 230 kV lightning-arrestor failure in Northern Virginia produced six voltage drops; approximately 1.5 GW of data-center load disconnected near-simultaneously. NERC's incident review concluded that the underlying mechanism scales with the size and clustering of the data-center fleet.³ ERCOT and East Coast operators have observed multiple events of this character in 2024–25. Synchronous generation cannot follow these ramps; the resulting voltage, frequency, and resonance interactions — including sub-synchronous oscillations — can damage equipment and threaten stability.⁴ As data centers running predominantly GPUs make up a growing share of new demand, we expect these types of events to increase without swift action.

ON urges the Commission to use this and related rulemakings to accelerate adoption of new standards for the interconnection of large computational loads to ensure grid reliability, power

³ NERC, *Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations*, 2025; NERC, *Aggregated Report on NERC Level 2 Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations*, 2026; NERC, *Incident Review: Considering Simultaneous Voltage-Sensitive Load Reductions*, Jan. 8, 2025; NERC, *Characteristics and Risks of Emerging Large Loads*, Reliability Standing Committee white paper, July 2025, available at www.nerc.com/pa/comp/CAOneStopShop/Pages/default.aspx

⁴ NERC, *White Paper: Characteristics and Risks of Emerging Large Loads* (July 2025), available at <https://www.nerc.com/globalassets/who-we-are/standing-committees/rstc/whitepaper-characteristics-and-risks-of-emerging-large-loads.pdf>.

quality, and system stability. The NERC alert provides a strong initial set of actions and should be codified into uniform, enforceable rules. ERCOT's large-load interconnection process for load facilities of 75 MW or greater, together with the pending NOGRR 282 and its associated work streams, establishes voltage and frequency ride-through requirements for Large Electronic Loads with explicit ramp-rate limits and addresses modeling, ramp behavior, operating coordination, and interconnection documentation. The principles in this ANOPR and the Essential Actions in the NERC Level 3 Alert are trying to solve the same problems. The Commission need not duplicate that work.

The risk of inaction is regional fragmentation: PJM, the California Independent System Operator (CAISO), the Midcontinent Independent System Operator (MISO), the Southwest Power Pool (SPP), and other grid operators each developing their own responses. Without a national baseline, a technical requirement in one region will mean something materially different than in another, making compliance monitoring, lessons learned, and equipment design needlessly complex. The NERC Level 3 Alert paired with the NOGRR 282 gives the Commission a ready-made national technical baseline for ensuring large computational loads support, rather than degrade, power quality and system stability. The remaining task is to make it binding and enforceable through this rulemaking.

Some stakeholders have raised, and will continue to raise, concerns that these rules will make it difficult, expensive, or impossible for data centers to interconnect. We disagree. Our equipment has been studied and approved for interconnection under ERCOT's interim large-load process pursuant to Senate Bill 6, and is designed to meet the pending Large Electronic Load ride-through requirements set out in NOGRR 282.⁵ The AI UPS™ is engineered to deliver the

⁵ ERCOT, *NOGRR 282 — Large Electronic Load Ride-Through Requirements*, Nodal Operating Guide Revision Request, submitted Nov. 14, 2025; status listed as "Pending" on ERCOT's issue page as of the filing date of these

deterministic ramp behavior, ride-through, and operator-controllable interface that NERC has identified as Essential Actions for reliable operation of the bulk power system. Its performance has been validated at the National Laboratory of the Rockies.⁶ We are currently deploying at data centers that needed a solution to meet ERCOT's interim requirements without needing to redesign their data center or delay their project. Our experience with other customers demonstrates that compliance can be achieved without significant delays or cost increases to the projects and, in some cases, can accelerate timelines and reduce project costs. Other technologies, such as parallel energy storage systems, flywheels, and more flexible IT equipment designs are also options to comply.

ON respectfully requests the Commission consider these comments in any rulemaking arising from this proceeding and accelerate adoption of standards governing large computational load interconnection to ensure system stability and power quality.

Respectfully Submitted,

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May 15, 2026

comments, available at <https://www.ercot.com/mktrules/issues/https://www.ercot.com/mktrules/issues/NOGRR282>. See also ERCOT, *Large Load Interconnection*, describing the interconnection process for load facilities of 75 MW or greater, available at <https://www.ercot.com/https://www.ercot.com/gridinfo/resource>

⁶ Caitlin McDermott-Murphy, *Could a New Kind of Power Supply Help Make Data Centers Grid-Friendly?*, National Lab. of the Rockies (Mar. 11, 2026), <https://www.nlr.gov/news/detail/program/2026/could-a-new-kind-of-power-supply-help-make-data-centers-grid-friendly>.